EA-193-40-18A Technical Data

Doc revision	Revision note	Date	Author
2	First release	20/03/2023	J.Seaton

Motor Type	Nominal Voltage	Rated Power
EA-193-40-18A	48 Vdc	5.5 kW

Performance Max Ratings			
Peak Torque	Peak Power	Max Phase Current	Maximum Speed
63 Nm	12.5kW	450 Arms	4500 RPM

Parameter	Symbol	Units	Value				Notes
Rated Performance (IEC Duty)							
Duty Type	-	-	S2-2min	S2-10min	S2-30min	S2-60min	EN IEC 60034-1
Torque	τ	*Nm*	63	27.2	17.8	13.3	"
Power	*P*	*kW*	6.6	10.0	6.5	5.6	"
Speed	*n*	*RPM*	1000	3500	3500	4000	"
Efficiency	η	%	69.3	91.5	95.1	95.3	"
Phase Current	I_{AC}	A_{RMS}	450	215	130.5	104.7	"
Phase Voltage	U_{AC}	V_{RMS}	15.2	33.6	30.9	31.7	"
Stator Frequency	*f*	*Hz*	83.3	292	292	333	"
Ambient Temperature	T_a	*°C*	-15 to 40	-15 to 40	-15 to 40	-15 to 40	"
Electrical Parameters							
Back EMF Constant	*ke*	V_{RMS}/*kRPM*	8				L-L Voltage
Torque Constant	*Kt*	*Nm/A*	0.15				
Stator Resistance	R_s	*mΩ*	5.01				At ambient 20°C
Inductance (d axis)	*Ld*	*µH*	36.7				Estimated
Inductance (q axis)	*Lq*	*µH*	67.5				Estimated
Pole Pairs	*PP*	#	5				
General Parameters							
Motor Weight	-	*kg*	11.4				Interface Dependant
Ingress Protection	*IP*		IP65, IP67 and IP69K options				EN IEC 60034-5
Material Thermal Class	-		220 (H)				EN IEC 60172
Motor Thermal Class	-		155 (F)				EN IEC 60034-1
Rotor Inertia	*Jr*	*kg mm2*	6109				Estimated
Ambient Limits	T_a	*°C*	-15 to 40				

Parameter	Value	Description	Notes
Position and Thermal Sensors			
Position Sensor	Sin/Cos	Sin/Cos output 1-4V	
Cycles per Revolution	5	One sin/cos cycle per electrical revolution	
Thermal Sensor Type	PT1000	Thermal sensor for stator winding	
Mechanical Interface			
Cooling Type (IC Code)	IC 410	Natural convection, no forced airflow	EN IEC 60034-6
Mounting Flange		See drawing for more detail	
Output Shaft		See drawing for more detail	

Customer Approval

Customer: Project:

Name: Signed: Date:

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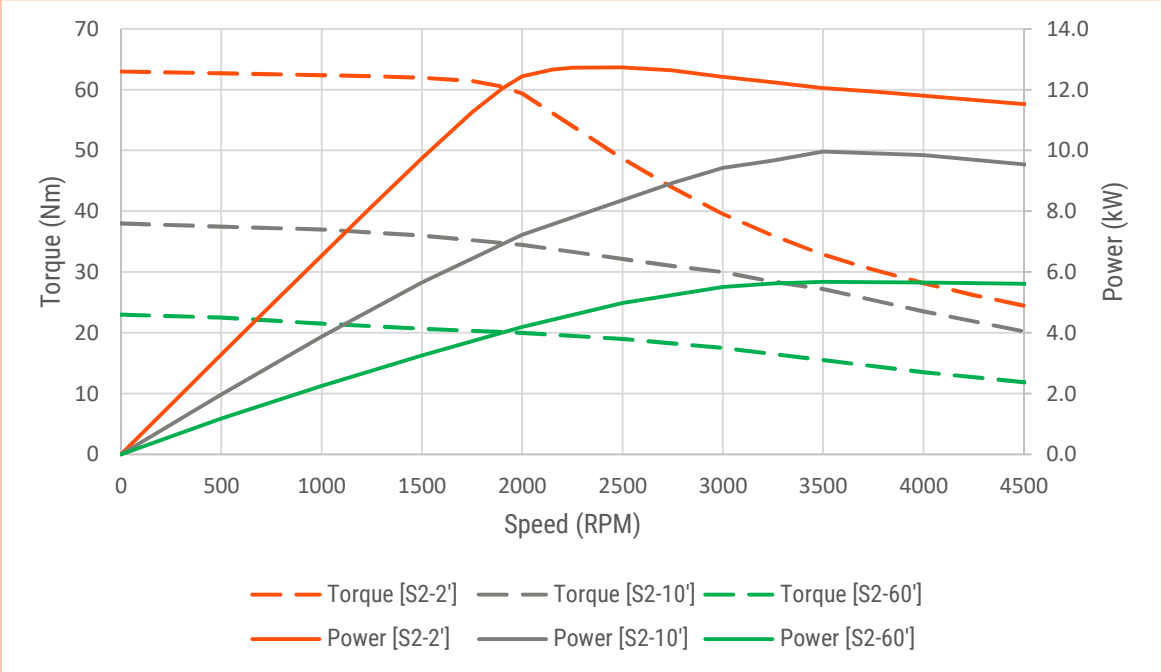
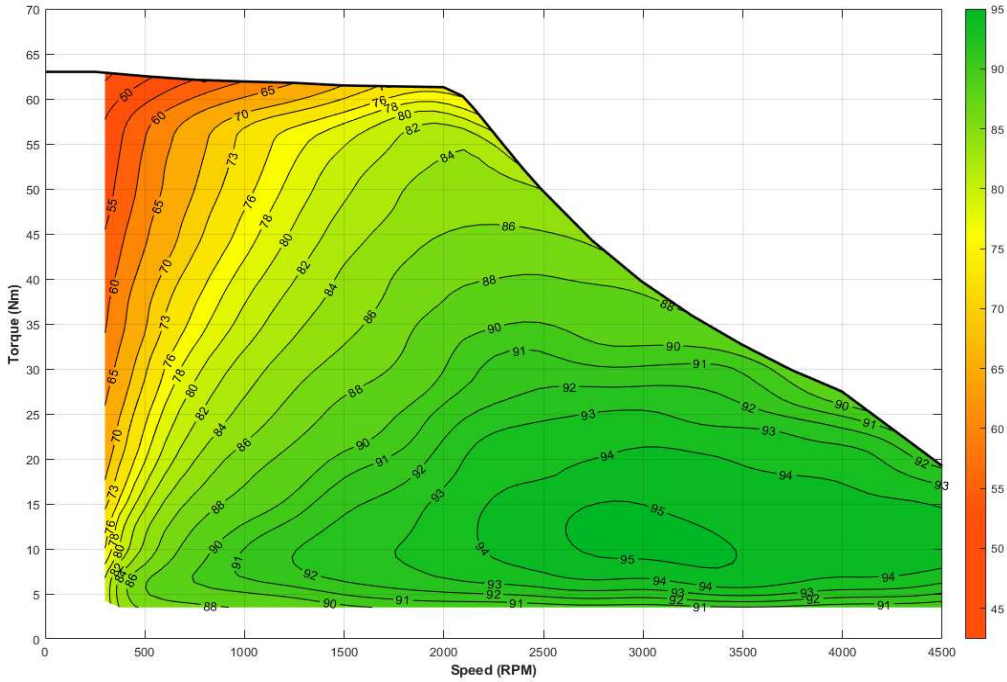
Torque and Power Curves



Motor Efficiency



NOTES:

1. Rated torque/power curves and efficiency map data measured at 20°C ambient.
2. All data is recorded with 48-50V nominal system voltage.
3. Performance ratings are based on typical operating conditions and optimised inverter settings.